

SARDAR PATEL UNIVERSITY
B.C.A (5th SEM) - EXAMINATION OCT 2025
US05DBCA21 - COMPUTER GRAPHICS

**Date: 08-10-2025(Wednesday)****Time: 10:00 AM to 01:00 PM****NOTE: Right hand figure indicates the marks of each question.****Total Marks: [70]****Q. 1 Multiple Choice Questions (One marks each)****[10]**

- 1 CRT stands for _____.
 (A) Cathode Ray Tube (B) Computer Ray Tube
 (C) Common Ray Tube (D) None of the above
- 2 Beam Penetration Method displays maximum _____ colors.
 (A) 8 (B) 4
 (C) 16 (D) 32
- 3 CAD stands for _____.
 (A) Computer Aided Design (B) Common Application Design
 (C) Common Aided Design (D) Computer Application Design
- 4 _____ rule (s) used to identify interior region of an object.
 (A) ODD – EVEN Rule (B) Non Zero Winding no. rule
 (C) Polygon fill (D) Both (A) & (B)
- 5 _____ obtained by adding a filled semicircle to each butt cap.
 (A) Round Cap (B) Circle Cap
 (C) Projecting Square Cap (D) Semi-Circle Cap
- 6 DDA stands for _____.
 (A) Data Drawing Algorithm (B) Digital Drawing Algorithm
 (C) Digital Differential Analyzer (D) Digital Drawing Analyzer
- 7 _____ transformation that distorts the shape of an object.
 (A) Translation (B) Shear
 (C) Rotation (D) Scaling
- 8 Rotation Point is also known as _____.
 (A) Rotation Angle (B) Rotation Axis
 (C) Pivot Point (D) Translation Vector
- 9 _____ Mode application initiates data entry.
 (A) Request Mode (B) Event Mode
 (C) Sample Mode (D) None of above
- 10 Which device specifying a coordinate position(x,y)?
 (A) Locator Device (B) String Device
 (C) Stroke Device (D) Pick Device

Q. 2 Short Questions. (Attempt any ten)

- 1 Explain the terms (1) Resolution, (2) Pixel.
- 2 List primary components of CRT and state its functions.
- 3 What are disadvantages of Beam Penetration Method?
- 4 Give difference between Bitmap Font and Outline Font.
- 5 Write only procedure for Boundary Fill Algorithm.
- 6 What is inside-outside test? List various methods used for it.
- 7 Define the term Scaling. List all types of Scaling.
- 8 Define: Rotation and Translation.
- 9 What is Clipping? List out all clipping object.
- 10 What is a user model?
- 11 Define: Control Icons and Application icons.
- 12 List various input modes.

Q. 3 A] Explain the working of CRT in detail.

[05]

B] Write note on Keyboard and Mouse.

[05]

OR

Q. 3 A] What is Computer Graphics? Explain in brief a major application area of Computer Graphics.

[05]

B] Differentiate between Raster scan displays and Random scan displays.

[05]

Q. 4 A] Explain Non-Zero Winding number rule with example.

[05]

B] Write steps of Bresenham's Line Drawing Algorithm.

[05]

OR

Q. 4 A] Explain Odd-Even rule with example.

[05]

B] Describe DDA algorithm with example.

[05]

Q. 5 A] What is text clipping? List out all types and explain it with example.

[05]

B] Define geometric transformation. List all 2-D geometric transformation and explain any one in detail.

[05]

OR

Q. 5 A] Explain in brief Sutherland Hodgeman polygon clipping algorithm.

[05]

B] Explain window-to-viewport coordinate window transformation.

[05]

Q. 6 A] Explain in detail General considerations in structuring a user dialogue.

[05]

B] List and explain Logical classification of Input Devices in detail.

[05]

OR

Q. 6 A] List and explain Interactive picture-construction techniques.

[05]

B] Explain Polygon surfaces in detail.

[05]